

NIZHNY NOVGOROD STRIGINO, Russian Federation

WMO: 275530

Lat: 56.230N

Lon: 43.784E

Elev: 82

StdP: 100.34

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.2	(c) -23.0	(d) -29.2	(e) 0.3	(f) -26.1	(g) -26.0	(h) 0.4	(i) -22.6	(j) 8.4	(k) -3.2	(l) 7.6	(m) -4.6	(n) 1.3	(o) 310	(p) 0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.8	(c) 30.2	(d) 20.1	(e) 28.2	(f) 19.4	(g) 26.8	(h) 18.8	(i) 21.3	(j) 27.6	(k) 20.5	(l) 26.6	(m) 19.6	(n) 25.1	(o) 2.8	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.1	23.8	18.2	13.2	22.7	17.5	12.6	22.2	62.4	27.6	59.2	26.6	56.2	25.2	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.4	5.6	-30.0	32.7	3.3	2.0	-32.4	34.2	-34.3	35.3	-36.1	36.4	-38.5	37.9		
			-30.2	23.3	3.2	1.3	-32.5	24.3	-34.4	25.1	-36.2	25.8	-38.5	26.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-9.1	-8.7	-2.8	5.9	13.0	16.8	19.3	17.2	11.6	5.1	-1.7	-6.5
	DBStd	11.39	7.10	7.14	5.03	5.09	4.95	4.22	3.54	3.94	4.17	4.49	5.40	6.67
	HDD10.0	2737	593	522	396	143	27	2	0	1	29	162	351	511
	HDD18.3	4973	851	756	654	373	177	78	31	68	204	411	601	769
	CDD10.0	948	0	0	0	21	118	207	289	225	78	9	0	0
	CDD18.3	142	0	0	0	0	11	33	62	34	3	0	0	0
	CDH23.3	1414	0	0	0	2	130	321	599	338	24	0	0	0
	CDH26.7	398	0	0	0	0	28	85	176	108	2	0	0	0
(14) Wind	WSAvg	2.5	2.8	2.8	2.9	2.7	2.5	2.3	2.0	2.0	2.1	2.6	2.8	2.8
(15) Precipitation	PrecAvg	606	40	31	28	35	49	66	75	65	57	60	55	46
	PrecMax	822	81	78	86	86	146	141	206	162	141	164	99	91
	PrecMin	439	4	0	5	3	2	12	6	2	8	2	17	6
	PrecStd	105	19	15	18	18	32	30	43	34	27	38	20	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	3.6	11.5	23.1	29.7	31.9	33.8	33.2	26.1	19.1	11.1	6.0
		MCWB	2.2	1.9	6.0	13.0	18.6	19.7	20.3	20.3	17.1	12.8	9.1	5.4
	2%	DB	2.0	2.2	7.2	20.1	26.9	29.0	30.1	29.4	23.2	15.2	8.1	4.0
		MCWB	1.3	1.2	3.4	11.5	16.9	19.3	20.3	20.1	16.2	11.4	6.9	3.3
	5%	DB	1.1	1.4	5.2	17.2	24.2	27.0	28.5	26.9	21.0	13.1	6.3	2.2
		MCWB	0.5	0.6	2.2	10.2	15.5	18.7	19.9	19.0	15.1	10.4	5.3	1.5
	10%	DB	0.2	0.7	3.5	14.9	22.0	24.9	26.8	24.8	18.9	11.5	4.8	1.1
		MCWB	-0.4	0.1	1.4	8.9	14.3	17.8	19.5	18.0	14.1	9.4	4.0	0.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	2.3	6.6	13.8	19.4	22.0	23.1	22.1	18.5	13.8	9.5	5.4
		MCDB	2.7	2.9	10.4	20.1	27.4	27.9	29.0	29.4	23.5	16.9	11.0	5.7
	2%	WB	1.2	1.4	4.0	12.3	17.8	20.7	21.6	20.8	16.9	12.1	7.1	3.2
		MCDB	1.5	1.9	6.3	18.6	24.7	27.2	27.3	27.5	21.7	14.6	7.8	3.6
	5%	WB	0.5	0.7	2.6	10.8	16.6	19.6	20.7	19.7	15.7	10.8	5.4	1.6
		MCDB	0.9	1.2	4.4	16.6	22.6	25.2	26.6	25.3	19.9	13.1	6.1	2.0
	10%	WB	-0.5	-0.1	1.6	9.4	15.3	18.5	19.9	18.6	14.5	9.6	3.8	0.6
		MCDB	0.2	0.6	3.1	14.2	20.4	23.4	25.5	23.5	18.3	11.2	4.4	1.0
(35) Mean Daily Temperature Range	MDBR		6.1	7.4	8.0	9.7	11.7	10.9	10.8	10.5	9.1	6.3	4.5	5.1
	5% DB	MCDBR	4.3	5.3	8.9	13.7	14.8	13.3	13.9	14.4	12.9	9.4	5.9	4.1
		MCWBR	4.2	4.7	6.0	7.1	6.8	5.9	5.9	6.5	6.7	6.3	5.2	3.9
	5% WB	MCDBR	4.4	5.1	7.4	12.7	13.1	11.9	12.0	13.0	11.6	8.8	5.5	4.0
		MCWBR	4.4	4.9	5.4	7.0	6.4	6.0	5.6	6.3	6.5	6.3	5.1	4.0
(40) Clear-Sky Solar Irradiance	taub		0.287	0.322	0.355	0.409	0.386	0.380	0.408	0.428	0.374	0.356	0.314	0.285
	taud		2.085	2.046	2.041	2.123	2.332	2.377	2.316	2.241	2.371	2.313	2.203	2.063
	Ebn at Noon		580	705	784	801	853	863	827	778	774	679	551	470
	Edh at Noon		61	96	124	134	115	112	117	118	90	74	55	49
(44) All-Sky Solar Radiation	RadAvg		0.58	1.43	2.96	3.99	5.40	5.55	5.58	4.42	2.68	1.22	0.47	0.31
	RadStd		0.11	0.29	0.34	0.41	0.35	0.51	0.53	0.47	0.39	0.21	0.09	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.64	N/A	N/A	N/A	N/A	N/A	-135	-180	N/A
(47) Regional (6 neighbors)		+0.63	N/A	N/A	N/A	N/A	N/A	-132	-179	N/A

Nomenclature: See separate page